
CE 504: Slope Stability and Retaining Structures

Course Code	CE 504
Course Name	Slope Stability and Retaining Structures
L-T-P-C	2-1-0-3
Prerequisites	CE 302 - Geotechnical Engineering
Intended for	UG/PG
Distribution	Elective
Approval	9th Senate (May 2016)

Course Contents

- **Module I**

Earth pressure, Rankine and Coloumb Earth Pressure theory, stresses and pressures for different conditions, [7 Lectures]

- **Module II**

Slope stability, types of slopes, theories of slope stability, seepage effects, theory of infinite slope stability, stabilisation methods. [12 Lectures]

- **Module III**

Design of retaining structures: gravity, cantilever, semi-gravity. [8 Lectures]

- **Module IV**

Soil reinforcement, earth retaining structures, design and analysis. [8 Lectures]

- **Module V**

Lateral earth pressure in braced cuts, Design of various components, Stability of braced cuts, base heave and stability, yielding and settlement of ground surrounding excavation. Diaphragm walls, slurry support; Soil Nailing. [7 Lectures]

Text Books

1. B. M. Das, **Principles of Foundation Engineering**, Cengage learning, 2011.
2. J. E. Bowles, **Foundation Analysis & Design**, 7McGraw-Hill Higher Education, 2001.
3. V. N. S. Murty, **Geotechnical Engineering**, CRC press, 2002.

References

1. S Hansbo, **Geotechnical Engineering**, Elsevier, 1994.
2. Hsai-Yang Fang, **Foundation engineering handbook**, Springer Science, 1991.